NIHVA

NIHVA Technologies Pvt. Ltd.

JIELONG

JIE LONG SHUTTER MOTORS LTD.

ECR Electronic Rolling Shutter Motor

Operating Manual



JIELONG SHUTTER MOTORS LTD

Warning

Read this manual carefully before the installation. Installation, testing and maintenance must be carried out by specially qualified personnel.

Periodical maintenance should take place in order to ensure smooth operation. If you have any questions feel free to contact our company.

- Our company has got certification by ISO9001-2000 internal quality control system.
- THE PEOPLE'S INSURANCE COMPANY (GROUP) OF CHINA has accepted our products liability insurance.
- New model chain system patent number: ZL 2004 2 0087351.2
- Break-proof chain insurance equipment patent number: ZL99209951.X、ZL02240227.6

Safety notice:

- The roller shuttrer door motors should be installed horizontally with precision.
- The roller axle of the shutter door should be homocentric and horizontal.
- The motor must be satisfactorily earthed to prevent potential injury from shocks .The earthing connection bolts should be fixed to the chain wheel support board or electric appliance control box.
- The roller shutter should be free of any obstructions.
- The vertical hanging length of the chain must be adjusted within 3-6mm,adjustment should be made before hanging the shutter onto the roller axis.
- It is strictly forbidden to pull on the motor down lead.
- External power cord should be minimum 1mm diameter.
- Special attention should be given to protecting the motor from humidity and rain, to prevent short-circuiting.
- The switch box has to be installed on the dry wall and placed at a height of over 1.5 meters high, this is to ensure that children cannot operate the wall switch and remote controller.
- Stand under the door is forbidden when installing.
- It is strictly forbidden to allow anyone or anything pass under the operating door.
- It is strictly forbidden to install the motor in combustible or explosive areas.
- Please refer to part four and five part if you want to know more about the left mounting, you can also refer to the installation diagram.
- Add appropriate lubricating oil into the chain rollers after motor testing; continue to do so when required.

- The shutter motor must to be installed up to 2.5 meters above the ground or accessible passageway.

Friendly Reminder:

Please choose the safety equipment as follow if you install our shutter motor:

1. Safety brake: Guard against the shutter sudden fall, when the motor is operating.
2. Photocell: Force the motor stop and go up, if the shutter motor go down and some barriers under the door.

Prologue

ECR electronic rolling shutter motor is the product which our company has been producing for many years. It utilizes most advanced production techniques combined with the latest technologies available to provide reliability with diverse functionality. The product has credible quality, it is simple to install and has proven to suit a wide variety of applications.

The motors can be installed to both Left and Right sides.

1. Work Environment and Conditions

1、Working temperature：－15℃～45℃。

2、Relative working humidity：≤95%（40℃）。

3、Short-term working：

a、Continual Single-Phase operation should not exceed 7minutes

b、Continual Three-phase operation should not exceed 15 minutes

2. Product characteristic:

- Simple pleasing design of structure with strong power.
- Good drive/size ratio.
- Selected elements ensure safe working, and reliability.
- Low noise levels.
- Small oscillation.
- Lightweight, simple installation.
- Simple manual operation for example: if there is a power cut.
- Safety device in national patent of chain-break.

3. main specification and technical parameters for ECR motor:

model	Input power (W)	Set load out put torque (N.m)	Set load lifting force (kg.f)	Set load out put turn speed (r/min)	Max lifting height (m)		Max external diameter of door (m)	Chain number	Weight of motor (kg)
					Slats thickness (mm)				
					12	20			
230V 50Hz									
84-1P	600	84	200	5.0	4	4	0.33	06B	11
168-1P	600	168	300	5.0	5	6	0.38	10A	11.6
225-1P	320	225	400	4.2	6	6	0.41	08B	7.9
343-1P	690	343	500	4.8	6	6	0.38	10A	12.2
412-1P	710	412	600	4.8	6	6	0.38	10A	12.2
647-1P	920	647	800	4.0	7	6.7	0.42	10A	14.2
809-1P	1013	809	1000	3.5	7	8	0.47	10A	14.2
1617-1P	1768	1617	1500	3.3	12	7.9	0.48	12A	28
380V 50Hz									
168-3P	460	168	300	5.0	5	6	0.38	10A	11.6
225-3P	250	225	400	4.2	6	6	0.41	08B	7.9
343-3P	490	343	500	4.8	6	6	0.38	10A	12.2
412-3P	490	412	600	4.8	6	6	0.38	10A	12.2
647-3P	750	647	800	4.0	7	6.7	0.42	10A	14.2
809-3P	780	809	1000	3.5	7	8	0.47	10A	14.2
1401-3P	1050	1401	1300	4.4	9.5	7	0.45	12A	23.5
1617-3P	1250	1617	1500	5.2	12	7.9	0.48	16A	23.5
2450-3P	1350	2450	2000	3.6	12	12	0.65	16A	27.5

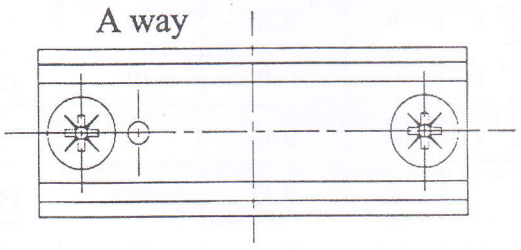
4. Installation and Testing:

ECR(300-1000) electronic rolling shutter motor is produced as standard for right side installation (viewed from inside). If you wish to install to the left side, it is necessary to loosen four bolts on the brake-shell. This will allow you to turn the shell 180° and then re-tighten the bolts

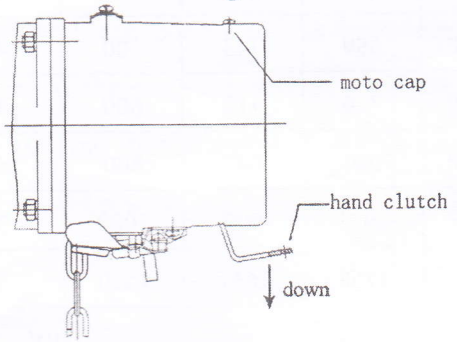
Note: The white and green lines of the switches should change positions.

ECR(1300-2000) electronic rolling shutter motor is produced as standard for right side installation (viewed from inside). If you wish to install to the left side, it is necessary to loosen four bolts on the brake-shell. This will allow you to turn the shell 180° and then tighten the bolts again. The white(up) and green(down) lines of the switches should change position. Turn the pawl on the shell 180 and retighten it(drawing 01). Hand chain can only move up the shutter door when there is a power cut. Because of the pawl, you can' t use the hand chain to move down the shutter door when power off, you can operate the hand hook on the brake-shell to move down the shutter door. Please refer to the instructions in part five(drawing 02).

drawing 01 one-way pawl



drawing 02 hand clutch



The support board of the chain-wheel can be installed horizontally, according to the specific situation as shown in Fig.3, or it can be fixed within a range of 0-45 degrees. When the support board is not installed horizontally the brake-shell must be turned so that the loop-type hand-pull chain can lie vertically in a normal manner.

Note: Special attention must be given to the joint of the shell-body and the top cover of the motor. When you are certain that there is no gap then tighten the bolts (diagonally). If a gap exists, and is allowed, the braking efficiency could be affected.

The phase order must not be violated for three-phase motors. When the rise-up becomes downside, the motors must be immediately stopped for readjustment of the phase order .If an interim power source is used for testing; attention should be given to connect in correct sequence when normal power source is introduced.

The correct operation approach is as follows: when the rise-up button on the White line is pressed, the motor should turn anti-clockwise, allowing the position limiting device to function. See Fig .1 for details. Otherwise the shutter door would be out of control and will be damaged.

Testing:

Screw cap position limiting slide piece model (Fig 2.1)

Loosen the Locking Bolt (No.3) of position limiter before testing, and then pull the hand chain to lift the door 1 meter above the floor. Press 'up' 'stop' and 'down' button, observe these functions of rolling shutter operation to see if they are correct. After you have confirmed that the door is working well, then open the door up or down to the position that you desire, turn the position limiter slide piece (No.6) until it touches joggling switch (No.2) and you hear 'ticktack'. Screw tight the Locking Bolt (No.3), and repeat the testing until the limiter is at the best position. Finally, screw the locking bolt tight again.

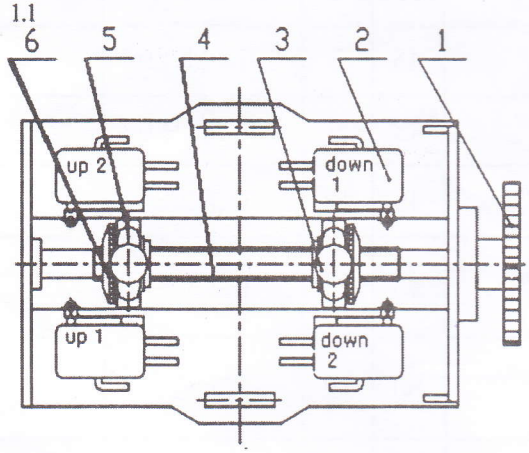
Gear form screw nut position limiting board model (Fig 2.2)

Loosen the two locking bolts (No.3) of position limiter, before testing. Hold the middle of position limiter board (No.6) with your fingers, push it 4mm to the left side? , and pull away the ?screw 4mm. The position limiting board will be separated from the gear form screw now. Turn the gear form screw nut (No.5) to set the door position, the method is the same as above. After setting the door position, hold the middle of position limiting board (No.6) and push it back up ? to make the position limiting board infix gear backlash. (If it cannot infix, turn the gear form screw nut to fit it), and then push the position limiting board to the right side ? 4mm, so as to reach the bottom of gear backlash. Screw tight the Locking Bolt (No.3), and repeat the testing until the limiter reaches the desired position.

If the shutter motor install debug or operate long time, the inside motor will heat and the thermal protection will force the motor stop operating. Till the temperature variation fall below 2K between the motor and the cooling medium, the motor will restart operate.

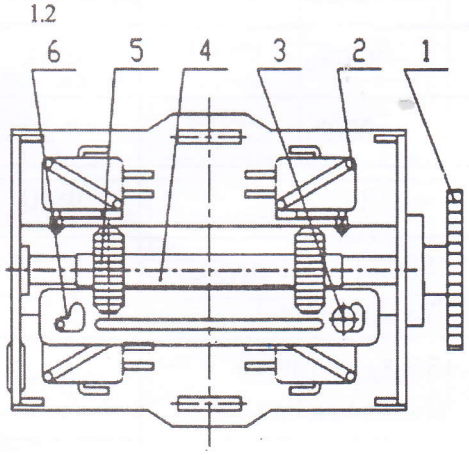
Fig 1 **Position Limiting Diagram**

1.1



1. gear 2. joggling switch 3. locking bolt
4. gearing wire 5.position limiting screw nut
6. position limiting slide piece

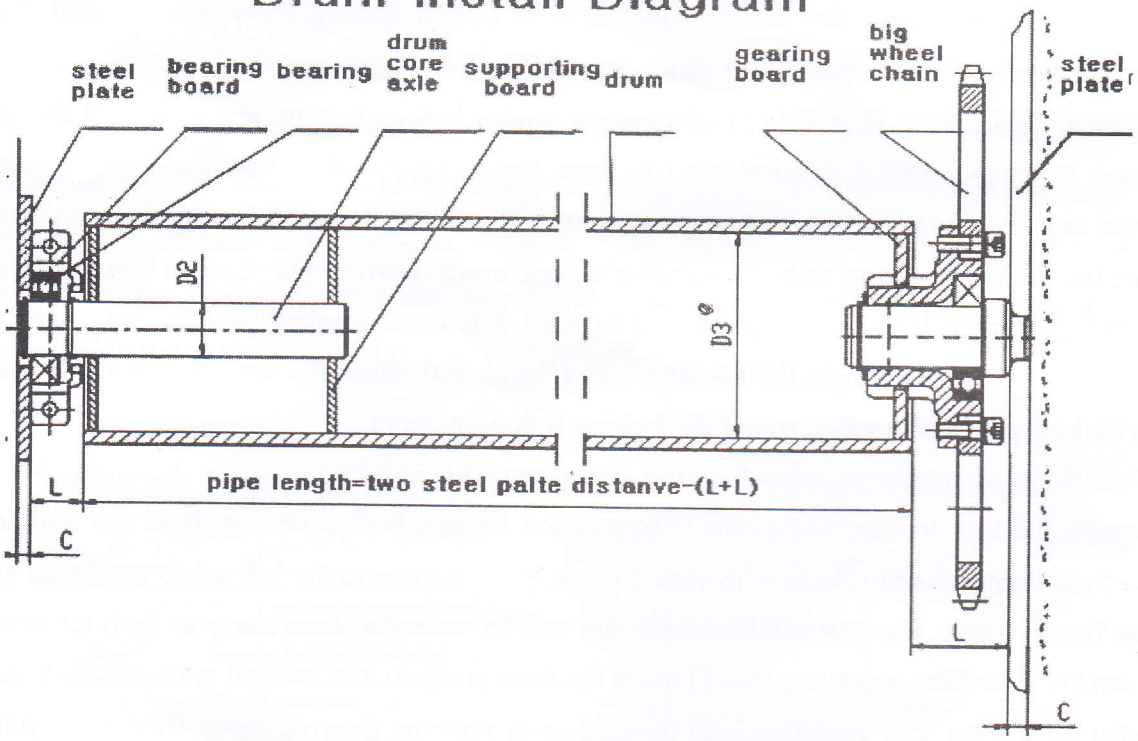
1.2



1. gear 2. joggling switch 3. locking bolt
4. gearing wire 5. gear form screw nut
6. position limiting board

Fig 2

Drum install Diagram



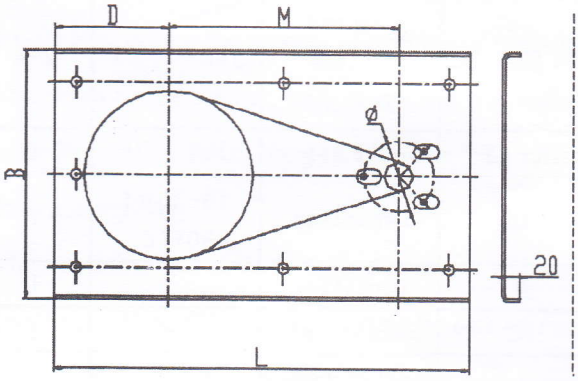
Drums install diagram data

model	D2	D3	L	C	Bearing	remark
200Kg	Φ30	3″ / 4″	35	3	206	
300Kg	Φ30	4″	35	4	206	
400Kg	Φ30	4″	35	3	206	
500Kg	Φ35	4″ / 5″	35	4	207	
600Kg	Φ35	5″	35	4	207	
800Kg	Φ35	6″	45	5	207and1207	
1000Kg	Φ40	6″	45	5	208and1208	
1300Kg	Φ40	8″	45	12	UCF208	
1500Kg (indoor installation)	Φ40	8″	55	12	UCF208	
1500Kg (outdoor installation)	Φ55	8″	80	12	UCF211	
2000Kg	Φ60	10″	85	16	UCF212	

Support plated installation diagram

Fig 3

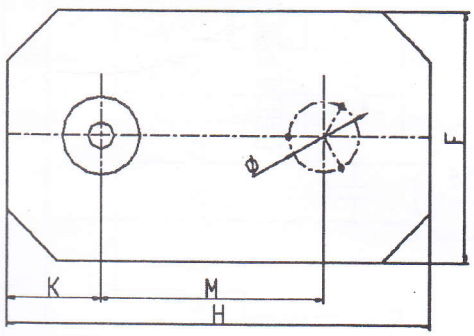
Model A



unit (mm)

Code no.	L	B	M	D	Φ
200Kg	450	220	264	100	
400kg	480	290	275	140	100
300-600kg	522	310	302	150	100
800kg	585	350	316	175	156
1000kg	620	400	352	185	156

General Model

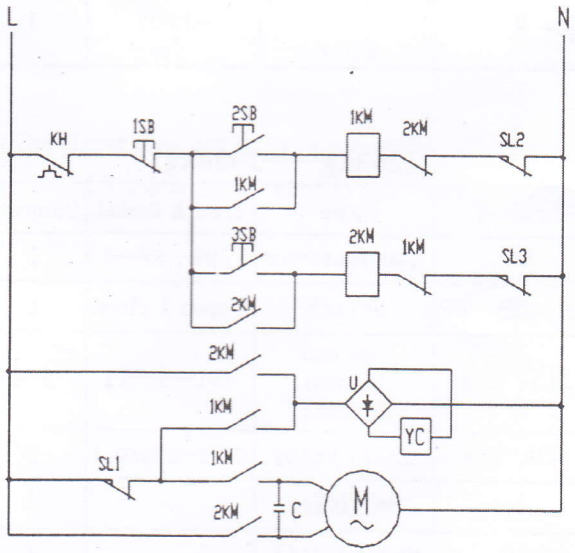


unit (mm)

Code no.	H	F	K	M	Φ
1300kg	520	200	80	347	186
1500kg	550	250	90	364	186
2000kg	650	250	96	454	186

Normally use horizontal installation

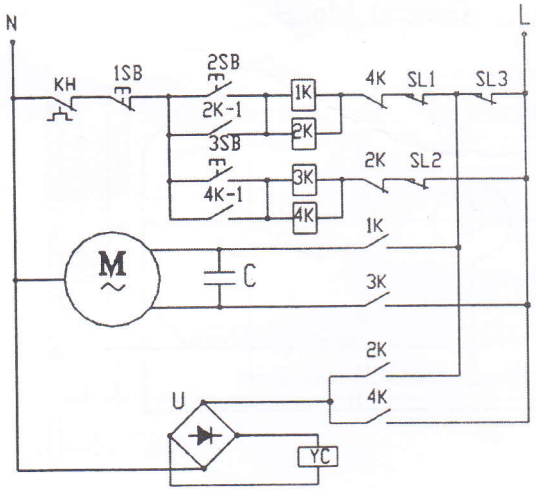
220V Electric Wiring Diagram



(300kg——1000kg)

Symbol	Name	Size & Model	Quantity
KH	Heat Protector	JOK—SF—1	1
1SB、2SB、3SB	Switch	2 open, 1 close	1
SL1、SL2、SL3	Top and bottom Buttons	JWL—1—11	3
1KM、2KM	Small relay	CJXZ—6.3/0.1	2
C	Capacitor	16-30μf 450VAC	1
U	Rectifier		1
YC	Magnet coil		1
M	Single—phase Motor	~220V	1

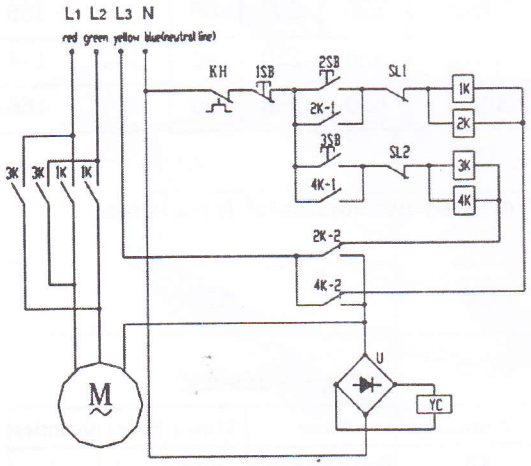
220V Electric Wiring Diagram



(300kg——1000kg)

Symbol	Name	Size & Model	Quantity
KH	Heat Protector	JOK—SF—1	1
1SB、2SB、3SB	Switch	2 open, 1 close	1
SL1、SL2、SL3	Top and bottom Buttons	JWL—1—11	2-4
1K—4K	Small relay	JQX—13F	4
C	Capacitor	16-30μf 450VAC	1
U	Rectifier		1
YC	Magnet coil		1
M	Single—phase Motor	~220V	1

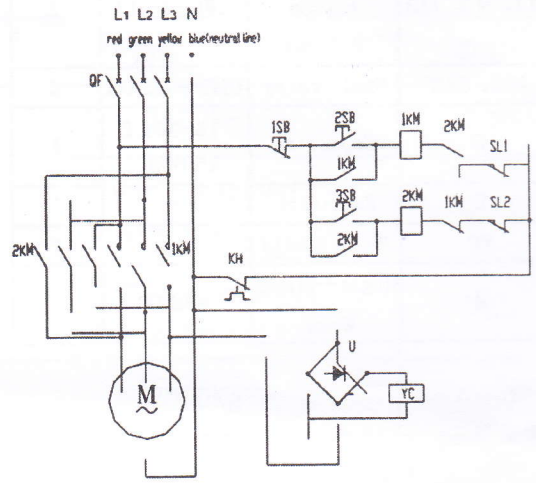
380V Electric Wiring Diagram



(300kg——1000kg)

Symbol	Name	Size & Model	Quantity
KH	Heat Protector	JOK—SF—1	1
1SB、2SB、3SB	Switch	2 open, 1 close	1
SL1、SL2	Top and bottom Buttons	JWL—1—11	2-4
1K—4K	Small relay	JQX—13F	4
U	Rectifier		1
YC	Magnet coil		1
M	Single—phase Motor	~380V	1

380V Electric Wiring Diagram



(1300kg——2000kg)

Symbol	Name	Size & Model	Quantity
KH	Heat Protector	JOK—SF—1	1
1SB、2SB、3SB	Switch	2 open, 1 close	1
SL1、SL2	Top and bottom Buttons	JWL—1—11	2-4
1KM、2KM	Small relay	CJXZ—6.3/0.1	2
U	Rectifier		1
YC	Magnet coil		1
M	Single—phase Motor	~380V	1

5. Usage

Press"up"or "down" buttons .If there is no reaction—press the "stop" button to avoid potential damage to the motor.

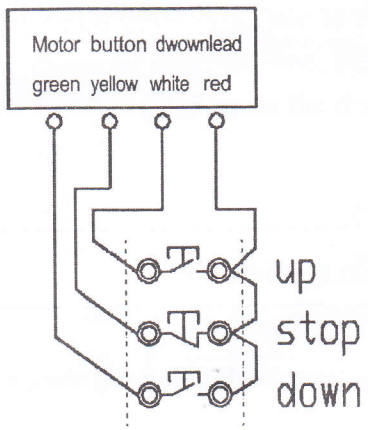
Check to see if there are any obstacles beneath the shutter door before closing it. Passage under the shutter is forbidden during the process of opening or closing the door.

When closing the shutter, with the power off, lightly pull on the shutter door loop for downwards, even, controlled speed. Relax the pull on the pulling loop when the door is almost closed and then pull again to close it completely.

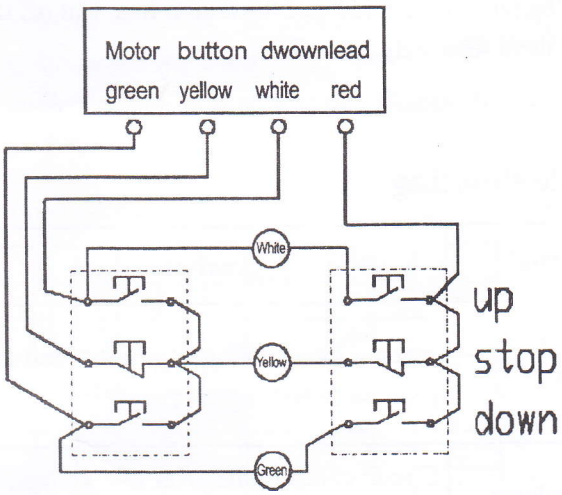
When opening the shutter, with the power off, strictly avoid putting the shutter over the height of the limiting position, ***Otherwise it may be out of control.***

Motors for fire shutter doors are equipped with separate control boxes. Specially qualified personnel should undertake regular and preventative maintenance and inspection.

Single-Switch Control

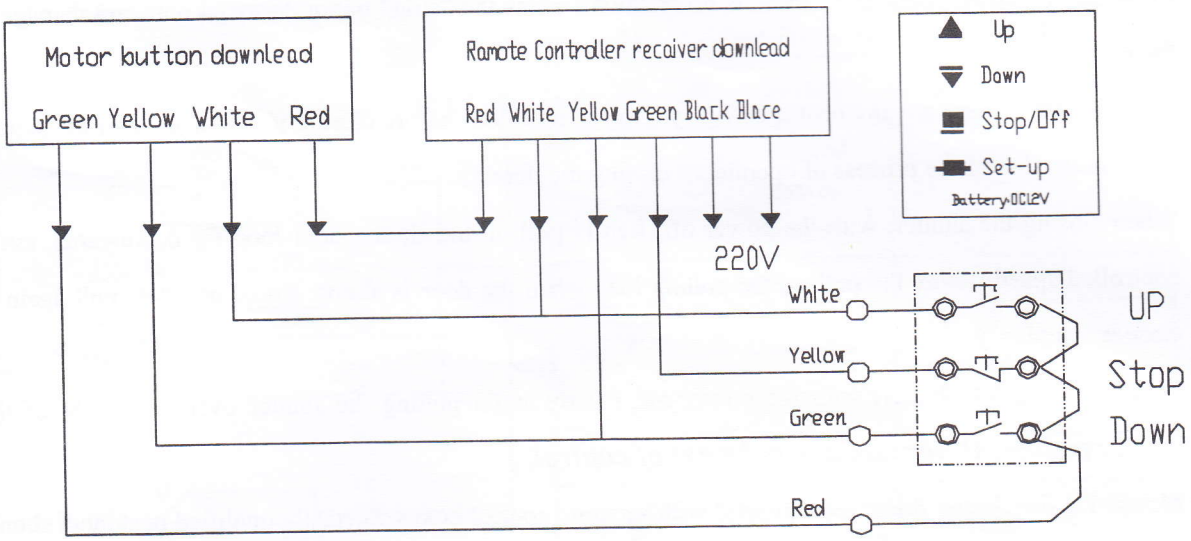


Double-Switch Dountead



☆The above diagram shows right-side installation. See Connection diagram
For left-side installation - change the position of the white and green lines.

Remote control and rolling shutter motor (button switch) connect diagram



☆See Connection diagram for right-side installation; change the position of white and green lines for left-side installation

☆Uncover the pipe at the middle of the button line, Cut off the end of the yellow line to uncover lead line, and connect them separately.

6. Trouble shooting

Problem	Analysis	Solution
No stop at upper and bottom positions	Incorrect connection of phases, and the position fails	Cut off power, check, adjust and change
Up only, or down only	Circuit contact fails, over low Voltage, Electromagnetic iron weak in attraction force	Polish contact or change relay, adjust voltage button contact, change magnet coil
No reaction	Button contact fails, position limiting switch contract fails, circuit cut off, without neutral wire connection	Adjust and change joggling switch and relay, connect neutral wire
Non-stop Operation	Relay contact fused, joggling switch out of control	Repair or change joggling switch and relay
Non-operation, only noise	Machine blocked, voltage too low	Remove block, adjust voltage
Shutter slide down after brake	Brake piece excess of abrasion	Check the spring of brake, add the shim.

7. Type Selection Formula

Total weight of shutter door G $\dfrac{\text{Output torque T}}{\text{Radius of rolling axle R}\times 9.8}$ **= Motor type –total lifted weight Kg**

G=door material/m^2×length ×width (see attachment for details)

material	aluminum alloy	net door	PVC steel	stainless steel	Fireproof door
KG×m²	5～6	12	8～14	10～18	25～36

Type and drum selection drawing:

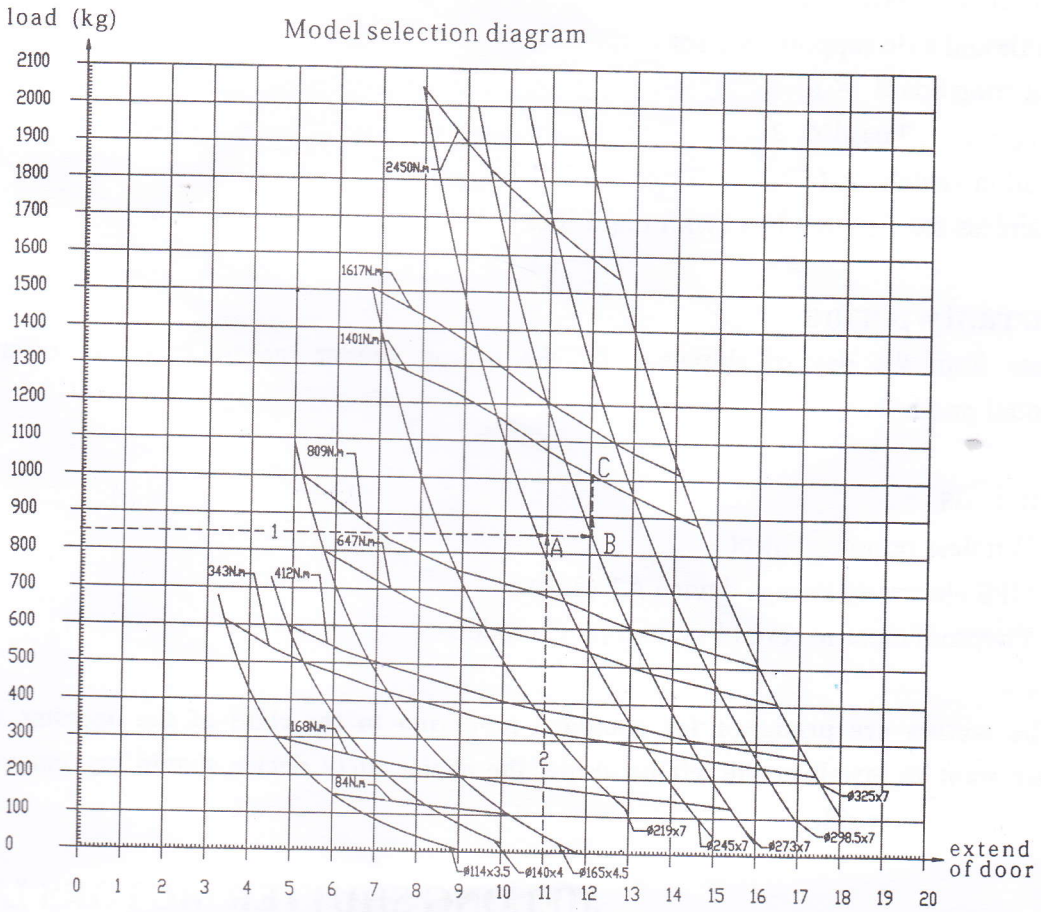
This diagram is drawn according to the national standard GB14102-2005 <fireproofing rolling shutter>, deflection =L/400, rolling shutter motor output torque, shutter gross weight and the span length.

Select method:

a) Single shutter type Selection weight G is actual weight ×(1.1~1.2);

b) Then according to the shutter Model Selection weight(kg), find out the relevant point from Y-coordinate, and to the across 1. According to the maximal width (m) of door to find out the relevant point from the X-coordinate, upwards to vertical line 2 to the Y-coordinate at across point A. Then along the right level direction, to reach the next point Bof pipe diameter diagonal line, that is the pipe diameter we want.

c) From point B up to the diagonal of type selection line is point C, that is the type of motor we want.



For example: the shutter maximal weight is 850kg and the door width is 11m, what size of pipe diameter and what kind of model shall we choose?

a) From Y-coordinate select 850kg as line 1. Then from X-coordinate select 11m as line 2 to reach the across of point A, along right direction to the next point B. The pipe diameter and thickness ? 273× 7 in pipe diameter line of point B, that is what we want.

b). From point B up to the closest model line at point C, the position of model 1401N.m is what we want.

- The deflection should be controlled less than 1/400, or when the door is more than 5m in width, with more than 3 mm thickness and self-weight is above 350kg, 5 seamless pipe with more than 3mm thickness and zinc plated must be used to prevent pipe axle from bending which would lead to overload and a short life of motor, and even directly damage teeth of motor's gear.
- Single-phase shutter door motor has obvious fluctuations of voltage, and therefore attention should be paid to voltage in type selection.

8. Accessories with motor

1. Chain wheel support ×1
2. Big chain wheel (includes axle connect) ×1
3. Rolling chain ×1
4. Drum core axle ×1
5. Axle and axle support ×1 set
6. Gearing board ×1
7. Supporting board× 2
8. Button switch ×1
9. Stainless steel switch box (with night lock) ×1

9. Warranty period

One year from the day of shipment for the rolling shutter motor. Parts are offered at preferential price.

10. Options

1. Wireless remote control
2. UPS electricity storage device AT (220)
3. Fireproof electric control system (A.C models);

Note:The motors are produced for standard mounting to the right of the opening, if the customer want to install to the left handside, the chain safety device should be remove and installed at the other side.

JIELONG SHUTTER MOTORS LTD